Fetal ECG Extraction by Blind Source Separation with Bio-Inspired Optimisation

How can bio-inspired optimisation algorithms be used to isolate the fetal cardiac activity from abdominal recordings in which it is mixed with the maternal ECG?

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Bio-Inspired Artificial Intelligence
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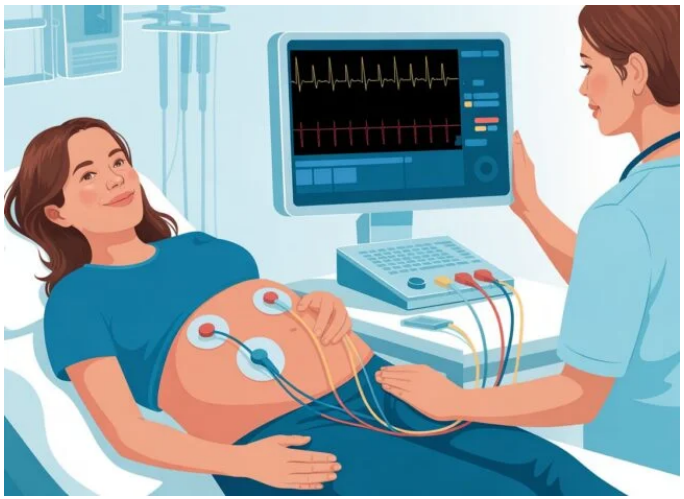
Context and related work

Topic

- Non-invasively measure the baby's vital signs, here its ECG, to support its development and safety
- The challenge: the fetal ECG is buried under the much stronger maternal ECG

Why it matters

- Precise fetal ECG today: invasive scalp electrode, during labour only
- Doppler cardiotocography: heart rate only, frequent signal loss
- This information is also valuable during a normal pregnancy



A happy mum: Owen and Morgan isolated her fetus's ECG [13]

Existing approaches

- Adaptive filtering (LMS, RLS): needs a thoracic maternal reference [1, 2]
- Maternal template subtraction [3]; Kalman filtering with an ECG model, single channel [4]
- Blind source separation (ICA, FastICA): multichannel, no reference [5, 6]; outperforms adaptive filtering [7]

Dataset

Database

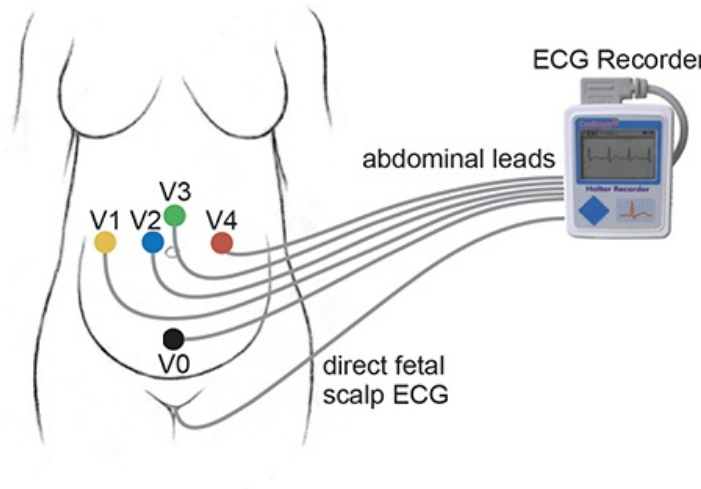
- Abdominal and Direct Fetal ECG Database, PhysioNet [9, 10]
- 5 women in labour (38–41 weeks), one 5-minute recording each

Recording conditions

- 4 electrodes around the navel, reference above the pubic symphysis
- Sampled at 1 kHz, band 1–150 Hz, power-line (50 Hz) and baseline drift removed

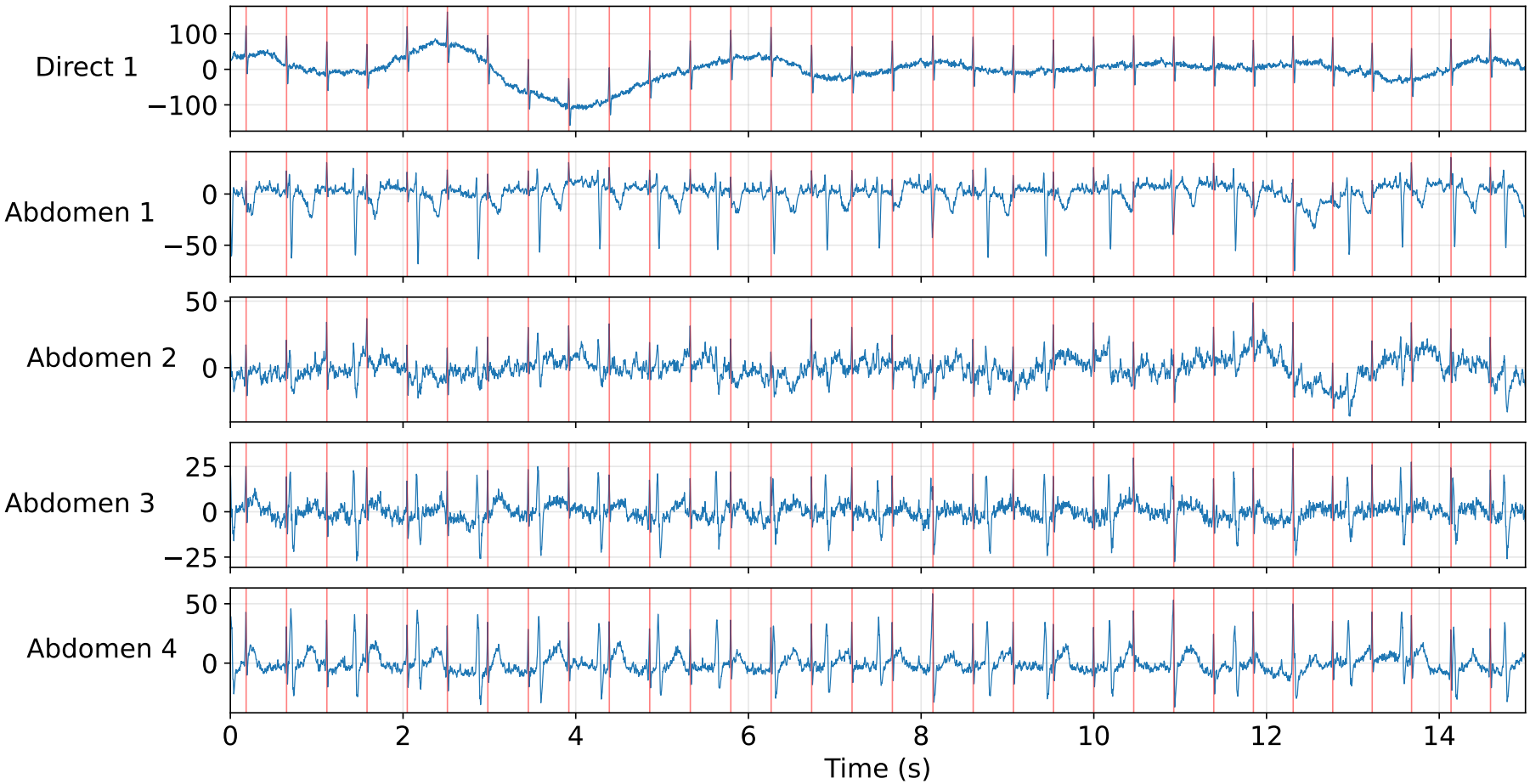
Available signals

- 4 signals from the maternal abdomen
- Direct fetal ECG from a scalp electrode, recorded simultaneously
- Fetal R-wave locations, verified by cardiologists



Example of electrode placement, not necessarily the exact setup of the database [8]

What the recordings look like



Record r01, first 15 s. Direct 1: scalp electrode. Red: annotated fetal QRS.

The optimisation problem

Mixing model, on one 15 s window ($N = 15\,000$ samples):

$$\underbrace{\begin{pmatrix} — \ \mathbf{x}_1 \ — \\ \vdots \\ — \ \mathbf{x}_4 \ — \end{pmatrix}}_{\mathbf{X}\ (4\times N)\text{: observed}} = \underbrace{\begin{pmatrix} h_{11} & \cdots & h_{1n} \\ \vdots & \ddots & \vdots \\ h_{41} & \cdots & h_{4n} \end{pmatrix}}_{\mathbf{H}\ (4\times n)\text{: unknown}} \underbrace{\begin{pmatrix} — \ \mathbf{s}_1 \ — \\ \vdots \\ — \ \mathbf{s}_n \ — \end{pmatrix}}_{\mathbf{S}\ (n\times N)\text{: unknown}} \quad \Longrightarrow \quad \mathbf{Y} = \mathbf{WX} \approx \mathbf{S}$$

BSS assumptions

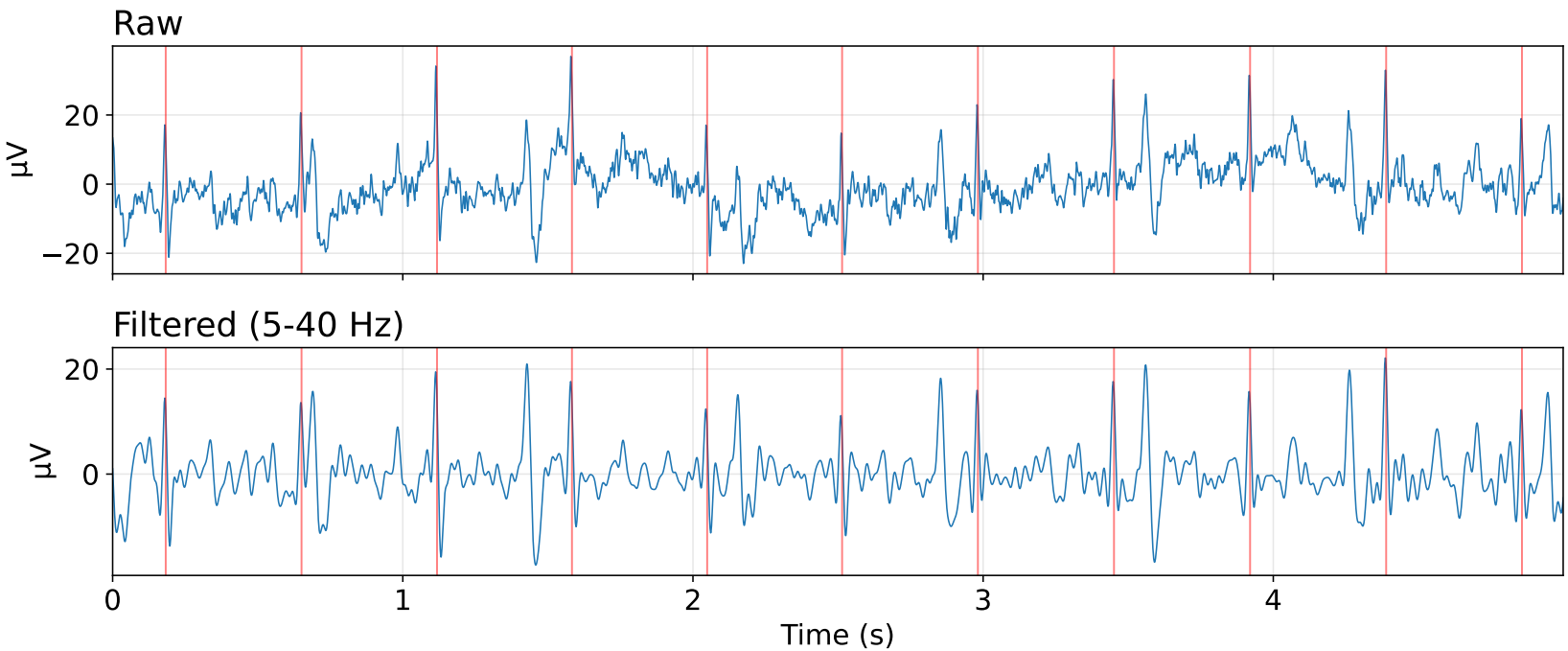
1. Instantaneous linear mixture
2. **H** constant: **15 s windows**
3. $n \leq 4$ sources
4. Independent, non-Gaussian sources

What we optimise

- **W** (4×4): 16 real numbers
- An independence criterion $J(\mathbf{W})$ on the outputs
- Candidates: mutual information, non-Gaussianity…
- Physiological constraints: periodic fetal output, 110–160 bpm

Preprocessing: filtering

- **Keep 5–40 Hz**, where the heartbeats are: removes slow drift (breathing, movements) and fast noise (muscles, mains)
- Same filter on every channel: the mixing model still holds



r01, Abdomen 2, first 5 s. Red: annotated fetal QRS.

Baseline: FastICA

Principle [6]

$$\underbrace{\mathbf{X}}_{\text{filtered, 15 s}} \longrightarrow \underbrace{\text{whitening}}_{\text{decorrelate}} \longrightarrow \underbrace{\text{fixed-point updates of } \mathbf{W}}_{\text{max. non-Gaussianity}} \longrightarrow \underbrace{\mathbf{Y}}_{\text{4 outputs}}$$

Why it is our reference

- Standard ICA algorithm; ICA is a classic approach for fetal ECG [5, 7]
- Fast, ready to use (scikit-learn)

Limits

- Local search: result depends on the initialisation
- One fixed criterion
- Fetal output must be identified afterwards

Beat detector

- ...

FastICA results

- ...

Next step: bio-inspired methods

Why bio-inspired

- No gradient needed: any criterion $J(\mathbf{W})$, including physiological constraints
- Population-based search, with restarts if needed: less dependent on a single initialisation
- Already explored for BSS: PSO [11], ACO with minimum mutual information [12]

Candidate methods

- Evolution strategies (e.g. CMA-ES), differential evolution, particle swarm optimisation
- One individual = one matrix $\mathbf{W}$ (16 real numbers)

Comparison with FastICA

- Same windows, same filtering
- Fetal beats detected vs. annotated fetal QRS
- Several runs: mean and spread, computation time

References I

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[2] S. Haykin, *Adaptive Filter Theory*, 4th ed., Prentice Hall, 2002.

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[8] R. Li, M. G. Frasch, H.-T. Wu, "Efficient fetal-maternal ECG signal separation from two channel maternal abdominal ECG via diffusion-based channel selection," *Front. Physiol.*, 8:277, 2017 (CC BY). Figure, slide 3: lead placement for the ADFECGDB.

[9] J. Jezewski, A. Matonia, T. Kupka, D. Roj, R. Czabanski, "Determination of the fetal heart rate from abdominal signals: evaluation of beat-to-beat accuracy in relation to the direct fetal electrocardiogram," *Biomed. Eng./Biomed. Tech.*, 57(5):383–394, 2012. Dataset: Abdominal and Direct Fetal ECG Database, PhysioNet, doi:10.13026/C2RP4B.

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References II

[12] N. Zhang, T. Liu, "The application of ant colony optimization algorithm in linear-combination blind source separation problem," *Proc. 2nd Int. Congress on Image and Signal Processing (CISP)*, 2009, doi:10.1109/CISP.2009.5304441.

[13] Eletrocardiograma.org, `https://eletrocardiograma.org/eletrocardiograma-in-pregnancy/`, accessed 11 November 2026. Illustration, slide 2.